

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 18

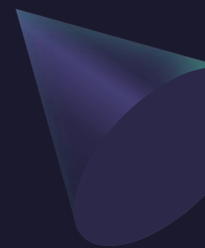


Any Reminders? Any Questions?

- Exam on Friday (discussion) covers 15.1-15.3 + 15.6-15.9
- See email message with resources
 - SP26 Practice Exam/Problems!
- Review during Thursday's class
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

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Review of Double and Triple Integrals



Example:

1. **Change of Order of Integration:** Consider the double integral of a continuous function $f(x, y)$ over the region D bounded by $y = x^2$, $y = 9$, and the y -axis in the first quadrant. Set up the integral in both orders: $dy dx$ and $dx dy$.



$$x^2=9 \Rightarrow x=3$$

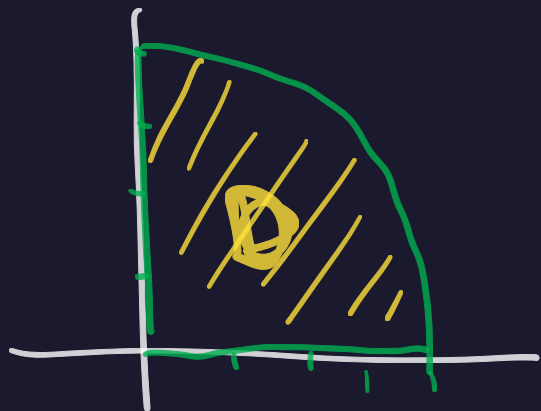
$$\iint_D f(x, y) dA$$
$$\iint_D f(x, y) dy dx$$
$$= \int_0^9 \int_0^{\sqrt{y}} f(x, y) dx dy$$

Example:

1. **Change of Order of Integration:** Consider the double integral of a continuous function $f(x, y)$ over the region D bounded by $y = x^2$, $y = 9$, and the y -axis in the first quadrant. Set up the integral in both orders: $dy dx$ and $dx dy$.

Example:

2. **Polar Coordinates:** Evaluate $\iint_D \frac{1}{\sqrt{x^2+y^2+1}} dA$ where D is the region in the first quadrant bounded by the circle $x^2 + y^2 = 16$ and the x and y axes.



$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$\frac{1}{\sqrt{x^2+y^2+1}} = \frac{1}{\sqrt{1+r^2}}$$

$$x^2 + y^2 = r^2$$

$$0 \leq r \leq 4$$

$$0 \leq \theta \leq \pi/2$$

$$\iint_D \frac{1}{\sqrt{x^2+y^2+1}} dA = \int_0^{\pi/2} \int_0^4 \frac{1}{\sqrt{1+r^2}} \cdot r \, dr \, d\theta$$

Example:

2. **Polar Coordinates:** Evaluate $\iint_D \frac{1}{\sqrt{x^2+y^2+1}} dA$ where D is the region in the first quadrant bounded by the circle $x^2 + y^2 = 16$ and the x and y axes.

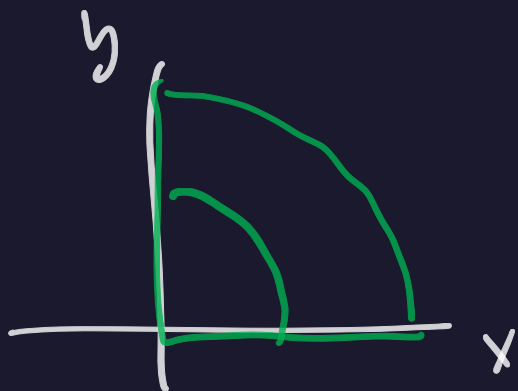
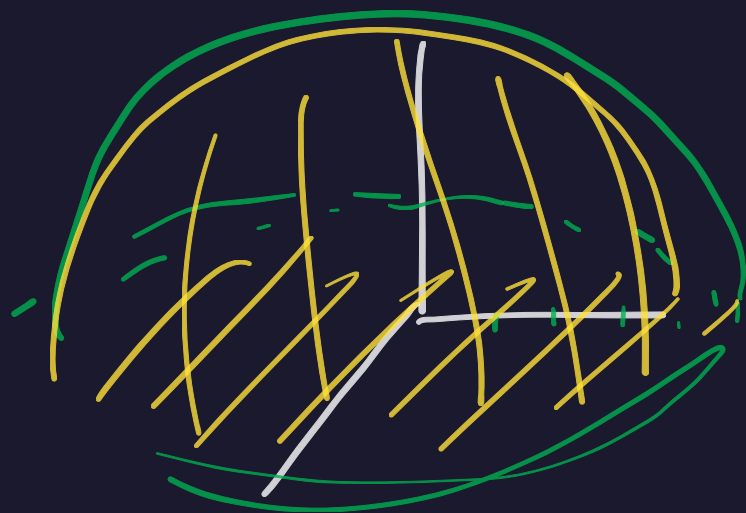
$$\int_0^{\pi/2} \int_0^4 \frac{r}{\sqrt{1+r^2}} dr d\theta \quad \begin{matrix} \text{u} = 1+r^2 & du = 2r dr \\ \downarrow \end{matrix} = \int_0^{\pi/2} \int_1^{17} \frac{1}{2} u^{-1/2} du d\theta$$

$$= \int_0^{\pi/2} u^{1/2} \Big|_1^{17} d\theta = \int_0^{\pi/2} (\sqrt{17} - 1) d\theta$$

$$= \boxed{(\sqrt{17} - 1) \cdot \frac{\pi}{2}}$$

Example:

3. **Spherical Geometry:** Find the volume of the solid E that lies within the sphere $x^2 + y^2 + z^2 = 25$ and above the xy -plane ($z \geq 0$). Set up the integral using spherical coordinates.

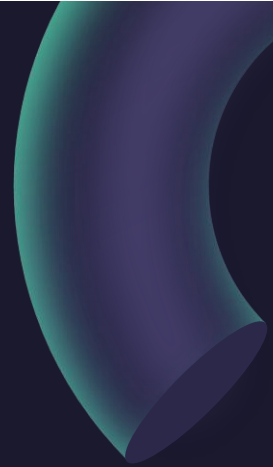
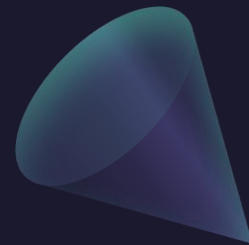


$$\int_0^{2\pi} \int_0^{\pi/2} \int_0^5 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$= \frac{250\pi}{3}$$

Example:

3. **Spherical Geometry:** Find the volume of the solid E that lies within the sphere $x^2 + y^2 + z^2 = 25$ and above the xy -plane ($z \geq 0$). Set up the integral using spherical coordinates.



Example:

4. **Jacobian Determinant:** Calculate $|J|$, the absolute value of the jacobian, for the transformation: $x = 4u + v, y = u - 2v$.

$$\begin{cases} x = 4u + v \\ y = u - 2v \end{cases}$$

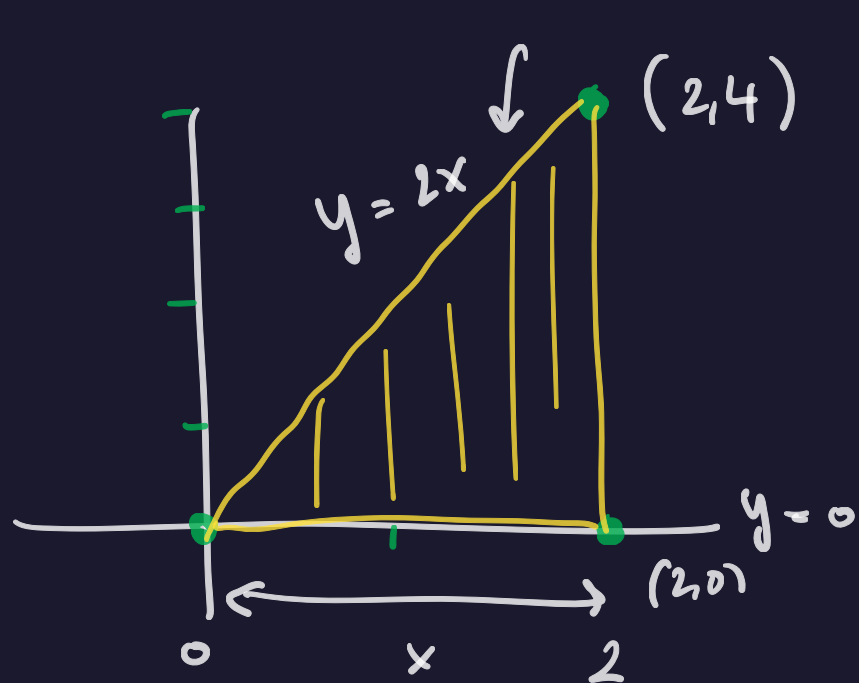
$$|J| = \left| \begin{vmatrix} 4 & 1 \\ 1 & -2 \end{vmatrix} \right| = |-8 - 1| = |-9| = 9$$

Example:

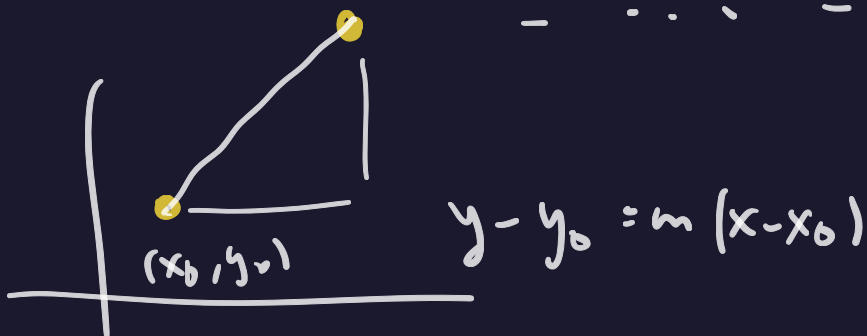
4. **Jacobian Determinant:** Calculate $|J|$, the absolute value of the jacobian, for the transformation: $x = 4u + v, y = u - 2v$.

Example:

5. **Evaluating Double Integrals:** Evaluate $\iint_D (3x - y) dA$, where D is the triangular region with vertices $(0,0)$, $(2,0)$, and $(2,4)$.



$$\int_0^2 \int_0^{2x} (3x - y) dy dx$$



$= \dots =$

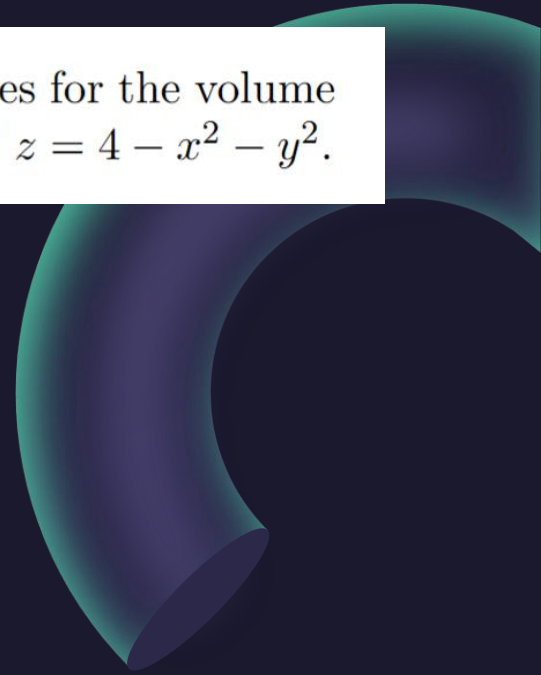
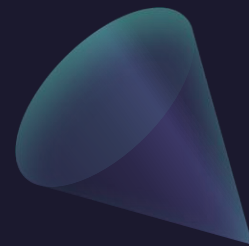
$$\boxed{\frac{32}{3}}$$

Example:

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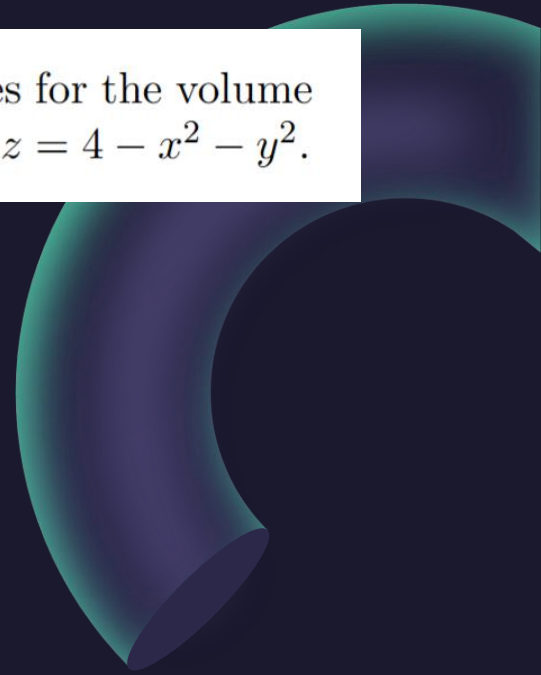
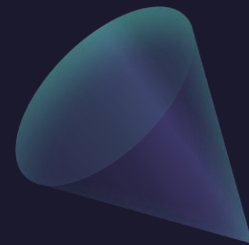
Example:

6. **Cylindrical Coordinates:** Set up a triple integral in cylindrical coordinates for the volume of the solid E bounded below by the xy -plane and above by the paraboloid $z = 4 - x^2 - y^2$.



Example:

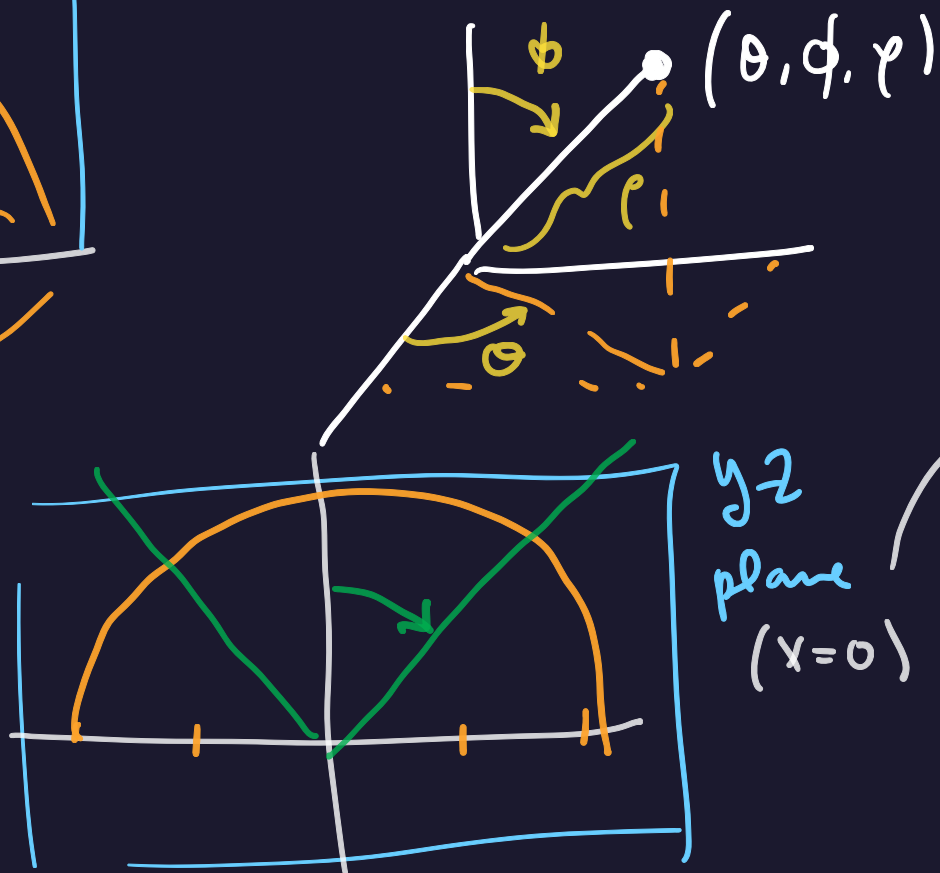
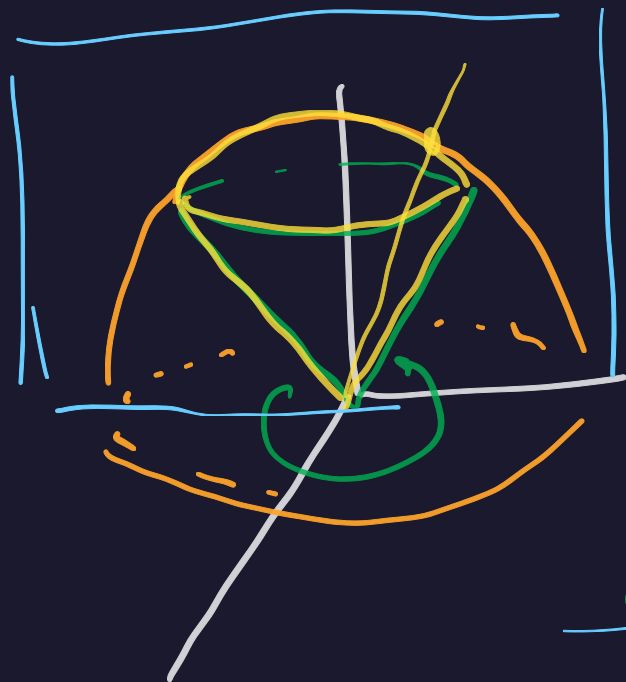
6. **Cylindrical Coordinates:** Set up a triple integral in cylindrical coordinates for the volume of the solid E bounded below by the xy -plane and above by the paraboloid $z = 4 - x^2 - y^2$.



Example:

7. **The Ice Cream Cone:** Let E be the solid region that lies above the cone $z = \sqrt{3(x^2 + y^2)}$ and inside the sphere $x^2 + y^2 + z^2 = 4$. Set up the integral $\iiint_E z^2 dV$ in spherical coordinates.

$$z = \sqrt{3(x^2 + y^2)} = \sqrt{3} \cdot \sqrt{x^2 + y^2}$$



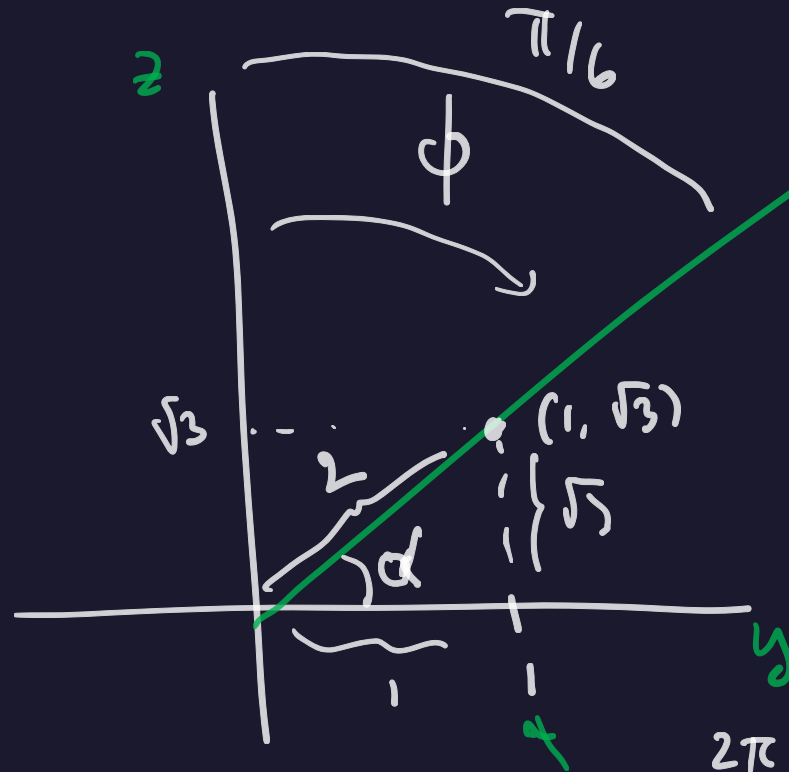
$$0 \leq \theta \leq 2\pi$$
$$0 \leq \rho \leq 2$$

$$y^2 + z^2 = 4$$
$$z = \sqrt{3}y^2 = \sqrt{3}|y|$$

$$z = \sqrt{3}y$$
$$z = -\sqrt{3}y$$

Example:

7. **The Ice Cream Cone:** Let E be the solid region that lies above the cone $z = \sqrt{3(x^2 + y^2)}$ and inside the sphere $x^2 + y^2 + z^2 = 4$. Set up the integral $\iiint_E z^2 dV$ in spherical coordinates.



$$z = \sqrt{3} y$$

$$\left. \begin{aligned} \sin \alpha &= \frac{\sqrt{3}}{2} \\ \cos \alpha &= \frac{1}{2} \end{aligned} \right\}$$

$$\begin{cases} x = \rho \sin \phi \cos \theta \\ y = \rho \sin \phi \sin \theta \\ z = \rho \cos \phi \end{cases} \quad \begin{cases} |\vec{r}| = \rho^2 \sin \phi \\ z^2 = \rho^2 \cos^2 \phi \end{cases}$$

$$\alpha = 60^\circ \Rightarrow \phi = 30^\circ = \frac{\pi}{6}$$

$$0 \leq \phi \leq \frac{\pi}{6}$$

$$\iiint_E z^2 dV = \int_0^{2\pi} \int_0^{\pi/6} \int_0^2 \underbrace{\rho^2 \cos^2 \phi \cdot \rho^2 \sin \phi}_{\rho^4 \cos^2 \phi \sin \phi} d\rho d\phi d\theta$$

Example:

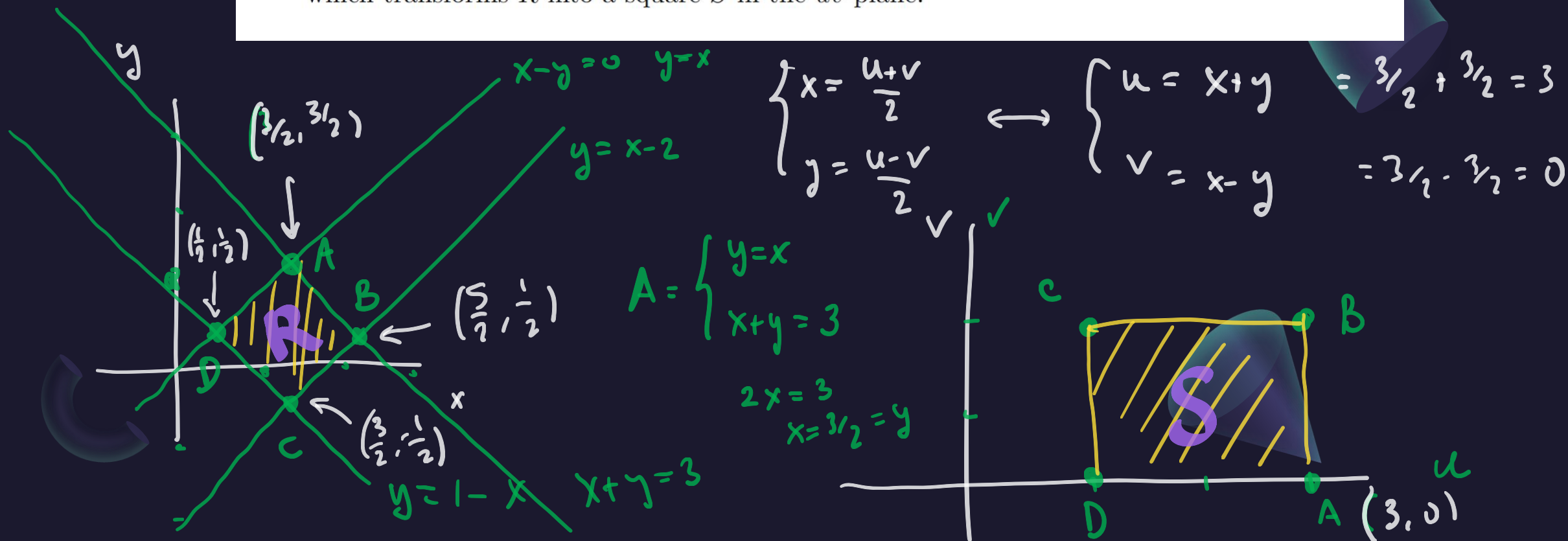
8. **Change of Variables:** Let R be the parallelogram in the xy -plane bounded by the lines

$$x + y = 1, \quad x + y = 3, \quad x - y = 0, \quad x - y = 2. \rightarrow y = x - 2$$

Evaluate the integral $\iint_R (x + y) dA$ by using the linear transformation

$$T(u, v) = (x, y), \quad x = \frac{u + v}{2}, \quad y = \frac{u - v}{2},$$

which transforms R into a square S in the uv -plane.



Example:

8. **Change of Variables:** Let R be the parallelogram in the xy -plane bounded by the lines

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which transforms R into a square S in the uv -plane.

$$x = \frac{1}{2}u + \frac{1}{2}v$$

$$\left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = \left| \frac{1}{4} - \frac{1}{4} \right| = \left| -\frac{1}{2} \right| = \frac{1}{2}$$

$$\iint_R (x + y) dA = \int_1^3 \int_0^2 \left(\frac{u+v}{2} + \frac{u-v}{2} \right) \cdot \frac{1}{2} dv du = 14$$

Example:

$$x = \frac{u+v}{2}$$

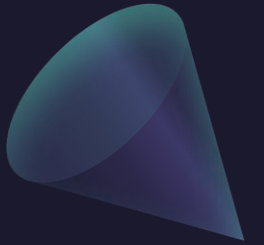
$$y = \frac{u-v}{2}$$

$$u = 2x - v \Rightarrow \boxed{u = 2x - (x - y) = x + y}$$

$$y = \frac{2x - v - v}{2}$$

$$\rightarrow y = x - v \Rightarrow \boxed{v = x - y}$$

Questions?



Thank you

Until next time.

